

Hints, Placebos and Shortcuts: Controlled Tests of Structural Primers for LLM Graph Reasoning

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Abstract

Prepending a *primer* of factual sentences about a graph’s structure (node degree, clustering coefficient, random-walk return probabilities or component count) to its text encoding is a natural way to assist a language model on graph-reasoning questions. An apparent gain is uninformative if the answer can be read from the primer without consulting the graph, and confounded if the primer merely lengthens the prompt. We evaluate primers against four controls jointly: a graph-blind shortcut solver refit at the evaluated graph size, a length-matched content-free filler, a minimal-length content control, and a degree-preserving rewiring that varies triangle count at fixed prompt length. Across four Qwen3 arms (1.7B and 4B, with and without thinking mode) on 40-node Erdős–Rényi graphs, primers that state the answer help a model at floor and harm a model at ceiling (cross-fitted interaction with baseline accuracy -11.0 points, 95% CI $[-17.1, -5.7]$), whereas primers that do not state it show no such relation. The pattern replicates on eleven held-out arms (0.6–14B, two families) but is not graded at intermediate baselines. The one primer that does not state the answer and still beats both controls, `clustering on node_degree`, gains $+5.9$ points at intermediate densities and little elsewhere; two candidate mechanisms for it are not supported.

1 Introduction

Fatemi et al. (2023) show that a large language model’s accuracy on graph-reasoning questions varies substantially with the choice of text encoding when the graph and question are held fixed. We ask a complementary question about the same benchmark: does prepending a short *primer* of derived, node-level facts to the graph encoding improve the model’s answers?

Four confounds complicate the interpretation of such an effect. First, a primer may help because

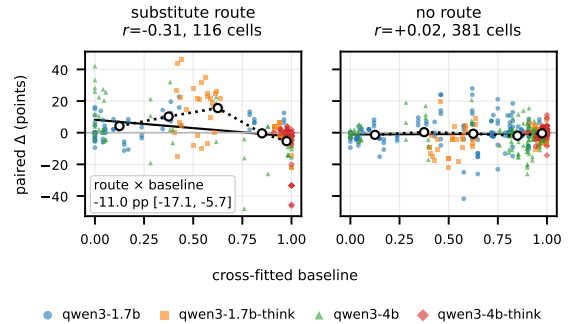


Figure 1: Cross-fitted paired effect against cross-fitted baseline accuracy (Section 5.3) for the 116 cells in which a primer offers a substitute route to the answer (left) and the 381 that do not (right), each shown once per fold direction. Baseline and effect come from disjoint halves of the paired graphs, so shared sampling noise cannot produce the slope.

the model reads the answer from it rather than reasoning over the graph (answer leakage). Second, a primer changes prompt length, and context length and position alone affect retrieval and reasoning accuracy (Liu et al., 2024; Kamradt, 2023; Levy et al., 2024). Third, answer extraction can err in ways that correlate with response style, which appears as a spurious condition effect (Section 3.4). Fourth, a model near ceiling on a task leaves no room for any primer effect to appear.

We address three research questions. **RQ1:** when a primer improves accuracy, is the improvement attributable to the primer stating the answer? **RQ2:** does added text affect accuracy independently of its content? **RQ3:** does any primer that does not state the answer improve accuracy beyond a length-matched control, and through what mechanism? Our main contribution is the four controls of Figure 2 applied jointly — a graph-blind shortcut solver refit at the evaluated graph size (Section 3.2), a length-matched `filler`, a minimal-length `components`, and a degree-preserving rewiring (Section 3.3) — on 40-node Erdős–Rényi

graphs across four model arms and six tasks.

2 Related Work

We adopt the GraphQA task definitions of [Fatemi et al. \(2023\)](#) and manipulate graph size and edge density directly, so that difficulty is an experimental variable rather than an incidental one. [Wang et al. \(2023\)](#) report that LLMs are brittle to spurious correlations in graph problems and propose *Build-a-Graph* and *Algorithmic* prompting, the prompt-level interventions closest to a primer; [Perozzi et al. \(2024\)](#) instead learn a continuous graph encoding. [Dziri et al. \(2023\)](#) argue that transformers solve compositional tasks by linearised subgraph matching rather than systematic reasoning, which raises the same question as a primer’s benefit: does it reflect reasoning or a shorter pattern to match? [Geirhos et al. \(2020\)](#) frame this as *shortcut learning*; our shortcut solver operationalises that concern for the primer channel. Irrelevant context degrades reasoning ([Shi et al., 2023](#)), as does input length well below the context limit ([Levy et al., 2024](#)); our *filler* condition measures this cost directly. The *rwse* condition adapts the random-walk structural encoding of [Dwivedi et al. \(2022\)](#), rendered as sentences, and the rewiring follows the double-edge-swap configuration model ([Fosdick et al., 2018](#)). We distinguish Qwen3’s native thinking mode, toggled through the chat template, from chain-of-thought *prompting* ([Wei et al., 2022](#)); this paper tests only the former.

3 Method

3.1 Primers

A primer is a preamble of factual sentences about a graph’s structure, prepended to the standard text encoding and the task question (Figure 2; a complete example appears in Appendix A.1). A single renderer produces all seven conditions, so they share sentence framing and number formatting and differ only in the facts stated. The exception is *filler*, which states no fact and therefore uses its own frame (“Node N is simply present within the graph G .”); it is a length control, not a content control. *rwse* states each node’s random-walk return probabilities after two and three steps ([Dwivedi et al., 2022](#)).

3.2 Shortcut solvers

A primer’s measured benefit is evidence of graph reasoning only if it exceeds what a solver that

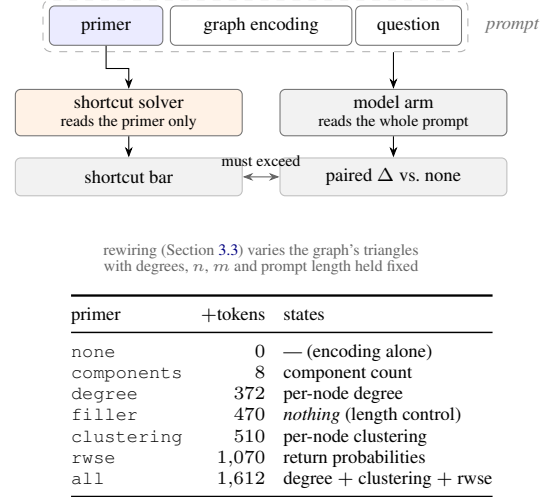


Figure 2: Experimental design. One prompt, two readers: the graph-blind shortcut solver sees only the primer and sets the bar a primer’s gain must exceed; the model arm sees all of it. Lengths are means at $n=40$ (Qwen3 tokenizer) against a mean *none* prompt of 2,165 tokens, so *filler* is a close length control for *clustering* but bounds only the length term for *rwse* and *all*.

never sees the graph can recover from the primer text. Our shortcut solver combines sixteen exact rules, one heuristic, eight fitted rules (fit and scored on disjoint graph sets) and an exact enumeration bound for small graphs. Its accuracy on a (task, condition) cell, the *shortcut bar*, is a content-only lower bound: it never overstates what the primer text alone permits.

Bars are fit at the evaluated graph size: $n=40$ Erdős–Rényi graphs at the seven densities of the sweep, with disjoint 300-graph fit and test sets per density (Table 1); the held-out arms, which answer the published 5–19-node split, use bars fit on that split, which differ substantially (Appendix Table 3). A (task, condition) pair offers a *substitute route* when its bar exceeds the *none* bar by more than 0.05; every gain is either at most 0.011 or at least 0.066, so any threshold between these values yields the same split. Only *node_degree* (under *rwse*, *degree*, *all*) and *edge_count* (under *clustering*, *degree*, *all*) offer a route. *node_count* and *cycle_check* have constant gold answers at $n=40$ (40, and “yes” at every density) and are excluded.

3.3 Rewiring: isolating structure from length

filler controls length only approximately, and comparisons across graph generators confound structure with class balance, so we use a within-

Task	none	comp.	clust.	rwse	degree	filler	all
node_count	1.00	1.00	1.00	1.00	1.00	1.00	1.00
edge_count	.030	.030	.104	.030	1.00	.030	1.00
node_degree	.144	.144	.144	.210	1.00	.144	1.00
connected_nodes	.000	.000	.000	.001	.001	.000	.008
edge_existence	.735	.735	.727	.554	.746	.735	.741
cycle_check	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Table 1: Shortcut bars at $n=40$ (mean over seven densities): accuracy of a solver that reads only the rendered primer, never the graph. **Bold:** the primer content alone determines the answer. The unbolded 1.00 entries reflect a constant gold answer, not primer content.

instance design. A double-edge swap replaces edges $(a, b), (c, d)$ with $(a, d), (c, b)$ (Fosdick et al., 2018), which preserves the degree sequence, n, m , the node_degree gold answer and the prompt length while changing the triangle count that the clustering primer reports. We generate low, base and high variants of each graph ($n=40$, mean degree 12, 200 graphs per level; mean clustering 0.01, 0.31, 0.73) and evaluate node_degree under none, clustering and filler with qwen3-1.7b, qwen3-1.7b-think and qwen35-2b. If the clustering benefit derives from the stated values, it should vary with triangle count. If instead the model reads mean clustering as a density prior (in an Erdős-Rényi graph $\bar{C} \approx p$), its degree estimates should shift toward $\bar{C}(n-1)$: 0.5, 12.0 and 28.7 across levels, against a true mean of 12.

3.4 Answer extraction and measurement validity

Extraction errors can masquerade as condition effects, so answers are extracted with task-specific patterns rather than a positional rule: on cycle_check a last-yes/no rule is not sufficient, because qwen3-4b often answers correctly and then recites that a cycle has “no repeated” nodes, which a positional rule scores against one condition more than another. A manual check of 50 rows per arm confirmed that <think> blocks never reach the extractor. Truncated pairs are dropped; scoring them as incorrect changes the verdict on only three qwen3-1.7b cycle_check cells (Appendix A).

3.5 Significance testing

Every comparison is paired on the same graph and question (Dror et al., 2018) and tested with the exact McNemar test (McNemar, 1947), Benjamini-Hochberg corrected (Benjamini and Hochberg,

1995) within each (arm, task) family at $q=0.05$; a claim that a condition beats both controls also requires significance under a global correction. For the effect-baseline relation (Section 5.3), baseline and effect are estimated from disjoint random halves of each cell’s graphs (Chernozhukov et al., 2018), and the route $\times$ baseline interaction is bootstrapped over (arm, task, density) blocks, which share one none sample. For null cells we simulate a minimum detectable effect (MDE) at power 0.8, $\alpha=0.05$.

4 Experimental Setup

Models and tasks. The six GraphQA tasks (Fatemi et al., 2023) are those named in Table 2. An arm is a model checkpoint with a fixed decoding configuration; the main arms are Qwen3-1.7B and Qwen3-4B (Qwen Team, 2025), each with native thinking mode off (qwen3-1.7b, qwen3-4b) and on (-think), which gives both modes of the same weights at two sizes.

Sweep design. All four arms answer one shared prompt set: Erdős-Rényi graphs (Erdős and Rényi, 1959) with $n=40$ at densities $p \in \{.10, .20, .35, .50\}$, 100 graphs per (density, task) cell, seven conditions and six tasks, for 16,800 prompts per arm. Decoding is greedy with a budget of 2,048 new tokens for plain arms and 8,192 for thinking arms; Section 5.5 reports a budget-matched comparison. A high-density extension covers node_degree and edge_existence at $p \in \{.65, .75, .85\}$ (100 graphs per cell, all four arms). For the one content effect that survives the controls (Section 5.4), a dedicated node_degree sweep with qwen3-1.7b uses 400 graphs per density at all seven densities, and a replication uses 400 newly seeded graphs per density at $p \leq .50$.

Held-out arms. Eleven further arms, held out but not pre-registered, were used for no claim reported here.¹

Metrics. Exact match for every task; mean absolute error for the integer tasks is in the Appendix.²

¹Eight (Qwen3-8B, Qwen3-14B, Gemma-4-E4B-it and Gemma-4-12B-it (Google DeepMind, 2026), each with and without thinking) answer the published GraphQA split (30 graphs per cell, five tasks, six conditions); three (Qwen3-0.6B with and without thinking, and Qwen3.5-2B (Qwen Team, 2026), qwen35-2b) answer a 100-graph probe of none and degree on all six tasks.

²Set-F1 on connected_nodes is 0.96–1.00 in every cell and does not discriminate between conditions.

5 Results

5.1 Main sweep

Table 2 gives the main sweep, and its largest entries are not primer effects. They fall on `node_count`, whose gold answer is the same for every graph, and on the two truncated `edge_count` cells, both flagged in the table. Across the four remaining tasks the entries are small and often change sign between the two model sizes. Two systematic patterns account for most of them: a cost of added text, which `filler` isolates (Section 5.2), and a benefit that appears only where the primer states the answer (Section 5.3). Both thinking arms answer nearly every question correctly under `none`, so the cells that can show an effect at all belong to the plain arms.

Throughout, Δ is a paired difference against `none` in percentage points, computed over instances in which neither condition was truncated; per-cell bootstrap intervals are in the Appendix. Outside the two flagged cells, every task truncates on fewer than 7% of instances.

5.2 Length is an active treatment

`filler` significantly reduces accuracy on `connected_nodes` and `edge_existence` in both plain arms, even though it states no fact about the graph. The cost grows with the difficulty of the task. Over the full seven-density range on `node_degree` it is harmless up to $p=.35$ and then costs between 9 and 14 points at the four densest levels, tracking accuracy under `none` across levels almost perfectly ($r=+.95$; 6.3 points pooled, $p<10^{-4}$). The cost of added text therefore grows in the same direction as the benefit of an answer-stating primer (Section 5.3).

The `edge_existence` effects are not changes in how the graph is read. The hit rate stays at 99–100% under every condition, so accuracy moves only through the false-alarm rate, which `filler` raises and `clustering` lowers. With hits at ceiling, sensitivity and criterion cannot be separated (Macmillan and Creelman, 2005; Hautus, 1995) (Appendix Tables 5 and 6). The `node_count` gains have a simpler explanation. The dominant error under `none` is an off-by-one, induced by the encoder’s “nodes 0, ..., 39” preamble and given by 68.5% of `qwen3-1.7b` answers, and almost any interposed text reduces it.

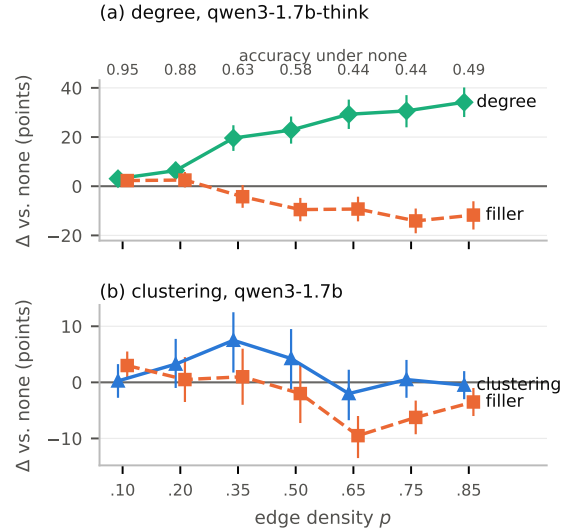


Figure 3: Paired effect against `none` on `node_degree` by edge density (400 graphs per density, 95% bootstrap intervals), on a shared density axis. (a) the answer-stating `degree` primer climbs as the task gets harder; (b) the non-answer `clustering` primer peaks at intermediate density. Note the $4\times$ difference in y scale between the panels: the content effect is an order of magnitude smaller than the leakage effect. `filler` is in both panels as the length control.

5.3 Answer-stating primers: route substitution

A *cell* is an (arm, task, density, condition) combination with at least ten usable pairs. On the main sweep plus the high-density extension, route cells gain where accuracy under `none` is low and lose where it is high, while no-route cells show no such relation (Figure 1; binned means also in Appendix Table 7). Baseline and effect share sampling noise, and a difference in significance is not a significant difference (Gelman and Stern, 2006), so we estimate the route $\times$ baseline interaction on cross-fitted halves of each cell’s graphs: -11.0 points per unit of baseline, with logit-scale and windowed robustness in Appendix Table 8. The relation is not graded. Restricted to cells whose baseline lies between 0.10 and 0.90 it disappears altogether, so what the data support is a contrast between floor and ceiling rather than a smooth dependence on competence: a stated answer helps a model that cannot compute the answer and hurts one that already can. Our coverage is thinnest exactly where a graded relation would have to show itself. Manipulating difficulty directly reproduces the contrast. For `qwen3-1.7b-think` on `node_degree`,

Arm	Task	none	comp.	clust.	rwse	degree	filler	all
qwen3-1.7b	connected_nodes	72.5	+0.8	+1.2	+1.0	+3.5	-6.2	+2.0
	cycle_check [†]	99.7	-0.6	+0.3	-7.5	-12.6	+0.3	-1.1
	edge_count	1.8	-1.4	-0.4	-0.8	+4.0	-0.4	+2.4
	edge_existence	70.5	-8.2	+3.8	+5.3	+4.0	-15.8	+11.5
	node_count [†]	31.5	+21.2	+67.2	+2.0	+6.8	+39.8	+47.0
	node_degree	60.5	+1.8	+3.0	+4.0	+7.5	-0.5	+10.2
qwen3-4b	connected_nodes	86.0	+9.2	+1.0	+0.0	-3.0	-12.5	-2.8
	cycle_check [†]	99.7	-1.3	+0.3	+0.3	-1.5	-0.3	-0.5
	edge_count	2.0	-1.3	-1.3	-1.5	+27.7	+1.2	+3.8
	edge_existence	90.5	-2.8	+4.0	-10.0	-4.3	-7.0	-9.8
	node_count [†]	90.2	+9.0	+8.8	+9.2	+8.0	+9.2	+9.8
	node_degree	99.2	+0.2	-0.5	-2.2	-6.5	-0.5	-8.0
qwen3-1.7b-think	connected_nodes	79.7	+2.8	+1.5	-0.3	+7.4	+2.0	+6.1
	cycle_check [†]	100.0	+0.0	+0.0	+0.0	-0.7	+0.0	+0.0
	edge_count [‡]	86.3	+7.4	-8.7	-4.0	-61.5	+3.4	-8.3
	edge_existence	99.2	-0.8	-1.0	-1.3	-2.3	-0.8	-2.0
	node_count [†]	98.6	+0.9	+1.4	+1.4	+0.8	+1.4	+1.4
	node_degree	76.8	-1.0	+1.0	+2.5	+11.7	-3.5	+11.6
qwen3-4b-think	connected_nodes	98.7	+0.5	+0.0	+0.0	+0.0	+0.8	-0.5
	cycle_check [†]	100.0	+0.0	+0.0	+0.0	-0.8	+0.0	+0.0
	edge_count [‡]	97.6	+1.7	+1.7	+0.0	-23.5	-2.0	-41.5
	edge_existence	100.0	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	node_count [†]	99.5	+0.0	+0.5	+0.5	+0.5	+0.5	+0.5
	node_degree	99.7	-0.3	-0.3	-2.1	+0.0	+0.0	-0.8

Table 2: Main sweep, $n=40$, pooled over densities $p \in \{0.10, 0.20, 0.35, 0.50\}$, 100 graphs per (task, primer, density) cell. The none column is absolute accuracy (%); all other columns are paired differences against none in percentage points, truncated generations excluded pairwise. [†]: gold answer is constant at $n=40$, so a shift need not reflect graph reading (Section 5.2). [‡]: $\geq 62\%$ of generations reach the output budget (Table 4). Both thinking arms sit at or near ceiling on most tasks, leaving no headroom for a primer effect. $n \in [13, 400]$ paired instances/cell. **Bold**: significant, exact McNemar, Benjamini–Hochberg within each (arm, task) family, $q = 0.05$.

raising density from $p=.10$ to $.85$ drives accuracy under none down and degree’s benefit up almost monotonically, from +3.1 to +34.2 points (Figure 3a).

Held-out arms and renaming. Three checks on the eleven held-out arms point the same way. The interaction has the same sign and direction there, though it is estimated far less precisely (Appendix Table 8). Substituting the other arms’ accuracy on a cell for the cell’s own baseline removes the relation, so it tracks a model’s own competence rather than the difficulty of the item. Replacing integer node identifiers with Game-of-Thrones names leaves degree’s effect essentially unchanged, so it does not depend on integer labels. None of this separates copying a stated value from computing the answer, but one sign of copying is that when a model errs despite a stated answer, the value it gives is its list-neighbour’s stated degree far more often than chance allows ($p=.002$).

5.4 Non-answer content: clustering on node_degree

clustering on node_degree is the only primer that does not state the answer and still exceeds both controls. This cell and the $p=.35-.50$ window below were chosen after the main sweep, so the cell was re-run with 400 graphs per density and an independent-seed replication that played no part in either choice (Figure 3b; Appendix Tables 9 and 10). Pooled over all seven densities the gain over none is small (+1.9 points), and the apparently larger gain over filler mostly reflects filler’s own cost, which accounts for the whole difference at the densest levels. The effect is concentrated in the middle of the density range. Pooled over $p=.35$ and $.50$, clustering exceeds none by +5.9 points (95% CI [2.1, 9.8]) and filler by a similar margin, while filler and none do not differ there, so length does not account for it. The replication gives +5.5 over the same two densities, and per-density estimates are too imprecise to localise the effect more finely. qwen3-4b is at ceiling in the main sweep but shows the effect on

the high-density extension, under the global correction against both controls; the thinking arms do not show it at all.

Neither mechanism we can test explains the effect. Under rewiring, mean clustering rises from 0.01 to 0.73 while the benefit stays flat, and for `qwen35-2b filler` matches or exceeds clustering at every level (Appendix Figure 4). The density-prior account fails as well. Signed error under clustering should shift by up to 17 at the high level, but the paired shift stays within 0.3 of zero at every level for all three arms (Appendix Table 11).

5.5 Thinking mode buys accuracy with budget

Thinking brings most cells to ceiling, and where it does not, truncation censors the task. Part of its advantage is the larger output budget rather than the reasoning it buys. Rescored at the plain arms’ 2,048-token budget, which is exact under greedy decoding, the 1.7B thinking arm still wins on `edge_existence` (97.8 against 70.5%) but loses on `cycle_check` (38.0 against 90.7%; Appendix Table 12).

Many of the main-sweep nulls are uninformative. Accuracy under `none` is already at floor or ceiling in a large share of the null cells of every arm, from 42% in `qwen3-1.7b` to all 33 in `qwen3-4b-think`, so no effect in at least one direction could be detected there at any sample size. Where a gain could be detected, the median detectable size is 5.6 points for the 1.7B arms and 10.0 for 4B.

6 Conclusion

Structural primers change LLM accuracy on GraphQA in two ways. For RQ1, a primer that states the requested quantity acts as a second route to the answer, helping a model at floor and harming one at ceiling; the contrast survives cross-fitting, a logit transformation, held-out arms, a negative-control difficulty estimate and node renaming, but is not graded at intermediate baselines. For RQ2, added text is an active treatment whose cost exceeds most content effects and grows with difficulty, so a primer compared only against `none` confounds content with length. For RQ3, the one surviving effect of non-answer content is small, concentrated at intermediate densities and unexplained. All claims are behavioural. Evaluations of prompt-level graph augmentations should report

baseline accuracy, a length-matched control and a graph-blind solver.

Limitations

The main sweep covers two model sizes from one family; the held-out arms extend the range but on a smaller corpus with fewer conditions and were not pre-registered. The route classification depends on a threshold and on the $n=40$ shortcut refit, and the route relation is not detectable within the 0.10–0.90 baseline window, where data are sparse. All graphs are Erdős–Rényi, in which local clustering is nearly uniform, so the clustering primer carries little node-specific information; the rewiring varies triangle count within this family but does not test other families. The length control matches clustering to within 8% in tokens but is much shorter than `rwse` and `all`. Every prompt uses one encoding (`incident`), one template and one primer position, with greedy decoding. Several results rest on 400 or fewer instances per cell; an MDE is reported for every null result, but it does not replace a larger sample. Three further primers would address the main open questions: *shuffled-value* (a stricter placebo than `filler`), *corrupted* (to separate copying from computation) and *single-fact* (to separate interference from route substitution).

Ethics Statement

This work evaluates open-weight language models on synthetically generated Erdős–Rényi graphs. It involves no human subjects, no personal or identifying data and no annotation labour. We see no direct risk of harm from the findings; the principal cost is the GPU compute for approximately 200,000 generations.

References

- Yoav Benjamini and Yosef Hochberg. 1995. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1):289–300.
- Victor Chernozhukov, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins. 2018. Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68.
- Rotem Dror, Gili Baumer, Segev Shlomov, and Roi Reichart. 2018. The hitchhiker’s guide to testing

- statistical significance in natural language processing. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 1383–1392.
- Vijay Prakash Dwivedi, Anh Tuan Luu, Thomas Laurent, Yoshua Bengio, and Xavier Bresson. 2022. Graph neural networks with learnable structural and positional representations. In *International Conference on Learning Representations (ICLR)*.
- Nouha Dziri, Ximing Lu, Melanie Sclar, Xiang Lorraine Li, Liwei Jiang, Bill Yuchen Lin, Peter West, Chandra Bhagavatula, Ronan Le Bras, Jena D. Hwang, Soumya Sanyal, Sean Welleck, Xiang Ren, Allyson Ettinger, Zaid Harchaoui, and Yejin Choi. 2023. Faith and fate: Limits of transformers on compositionality. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36.
- Paul Erdős and Alfréd Rényi. 1959. On random graphs i. *Publicationes Mathematicae Debrecen*, 6:290–297.
- Bahare Fatemi, Jonathan Halcrow, and Bryan Perozzi. 2023. Talk like a graph: Encoding graphs for large language models. *arXiv preprint arXiv:2310.04560*.
- Bailey K. Fosdick, Daniel B. Larremore, Joel Nishimura, and Johan Ugander. 2018. Configuring random graph models with fixed degree sequences. *SIAM Review*, 60(2):315–355.
- Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. 2020. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2(11):665–673.
- Andrew Gelman and Hal Stern. 2006. The difference between “significant” and “not significant” is not itself statistically significant. *The American Statistician*, 60(4):328–331.
- Google DeepMind. 2026. Gemma 4 model cards (gemma-4-E4B-it, gemma-4-12B-it). <https://huggingface.co/google>.
- Michael J. Hautus. 1995. Corrections for extreme proportions and their biasing effects on estimated values of d' . *Behavior Research Methods, Instruments, & Computers*, 27(1):46–51.
- Greg Kamradt. 2023. Needle in a haystack – pressure testing llms. https://github.com/gkamradt/LLMTest_NeedleInAHaystack.
- Mosh Levy, Alon Jacoby, and Yoav Goldberg. 2024. Same task, more tokens: the impact of input length on the reasoning performance of large language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 15339–15353.
- Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. 2024. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12:157–173.
- Neil A. Macmillan and C. Douglas Creelman. 2005. *Detection Theory: A User’s Guide*, 2nd edition. Lawrence Erlbaum Associates.
- Quinn McNemar. 1947. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2):153–157.
- Bryan Perozzi, Bahare Fatemi, Dustin Zelle, Anton Tsitsulin, Mehran Kazemi, Rami Al-Rfou, and Jonathan Halcrow. 2024. Let your graph do the talking: Encoding structured data for LLMs. *arXiv preprint arXiv:2402.05862*.
- Qwen Team. 2025. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*.
- Qwen Team. 2026. Qwen3.5-2B model card. <https://huggingface.co/Qwen/Qwen3.5-2B>.
- Freda Shi, Xinyun Chen, Kanishka Misra, Nathan Scales, David Dohan, Ed H. Chi, Nathanael Schärli, and Denny Zhou. 2023. Large language models can be easily distracted by irrelevant context. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, volume 202 of *PMLR*, pages 31210–31227.
- Heng Wang, Shangbin Feng, Tianxing He, Zhaoxuan Tan, Xiaochuang Han, and Yulia Tsvetkov. 2023. Can language models solve graph problems in natural language? In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. 2022. Chain-of-thought prompting elicits its reasoning in large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35.

A Supplementary material

A.1 Example prompt

A clustering-condition node_degree prompt at $p=0.10$ (elisions marked ...). Other conditions replace the first paragraph; none omits it.

Node 0 has clustering coefficient 0.20. Node 1 has clustering coefficient 0.00. ... Node 39 has clustering coefficient 0.00.

G describes a graph among nodes 0, 1, 2, ..., 38, and 39.

In this graph:

Node 0 is connected to nodes 2, 6, 8, 12, 35.

Node 1 is connected to nodes 5, 20.

...

Node 39 is connected to nodes 15, 16, 18, 20.

Q: What is the degree of node 16?

A:

A.2 Robustness analyses

Task	none	comp.	clust.	rwse	degree	filler	all
node_count	.064	.076	1.00	1.00	1.00	1.00	1.00
edge_count	.018	.018	.148	.018	1.00	.018	1.00
node_degree	.082	.082	.082	.616	1.00	.082	1.00
connected_nodes	.082	.082	.082	.082	.208	.082	.352
edge_existence	.498	.498	.720	.642	.794	.498	.794
cycle_check	.832	1.00	.832	.832	.946	.832	.946

Table 3: Shortcut bars on 500 graphs from the published 5–19-node generator, used for the held-out arms. At $n=40$ (Table 1) the `edge_existence/none` bar rises to 0.735 through class imbalance, and `degree`’s gap over `none` on that task falls from +0.296 to +0.011.

Task	1.7B	1.7B-th.	4B	4B-th.
edge_count	22.4%	81.0%	0.6%	62.2%
cycle_check	6.1%	6.5%	1.0%	1.9%
node_degree	0.0%	1.1%	0.0%	1.6%
connected_nodes	0.0%	0.8%	0.0%	3.5%
edge_existence	0.0%	0.2%	0.1%	0.8%
node_count	0.0%	2.5%	0.0%	0.1%

Table 4: Truncation rate (share of main-sweep generations reaching the output budget: 2,048 tokens plain, 8,192 thinking), by task and arm, pooled over condition and density. **Bold**: above 50%.

Arm	Cond.	hit rate	false alarm
qwen3-1.7b	none	100.0%	40.3%
	filler	100.0%	61.8%
	clustering	100.0%	35.2%
qwen3-4b	none	99.1%	12.6%
	filler	100.0%	22.5%
	clustering	100.0%	7.5%

Table 5: `edge_existence`, $p=0.50$: hit rate (correct on gold “yes”) and false-alarm rate (“yes” on gold “no”). Accuracy changes almost entirely through the false-alarm rate.

Cond.	qwen3-1.7b		qwen3-4b	
	$\Delta d'$	Δc	$\Delta d'$	Δc
components	−0.36 [−0.56, −0.21]	−0.18 +0.07 [−0.24, +0.33]	−0.10	
clustering	+0.21 [+0.03, +0.41]	+0.11 +0.43 [+0.12, +0.75]	+0.08	
rwse	+0.07 [−0.19, +0.43]	+0.28 −0.38 [−0.68, −0.15]	−0.32	
degree	+0.15 [−0.05, +0.41]	+0.19 −0.04 [−0.33, +0.21]	−0.15	
filler	−0.93 [−1.16, −0.71]	−0.46 −0.25 [−0.54, +0.00]	−0.25	
all	+0.55 [+0.35, +0.77]	+0.27 −0.36 [−0.71, −0.11]	−0.31	

Table 6: `edge_existence`, main sweep: change in sensitivity d' and criterion c against `none`, averaged over the four densities (log-linear correction; 95% bootstrap CI for $\Delta d'$). With the hit rate at ceiling, d' and c are not separately identifiable.

Accuracy under	substitute route		no route	
	none	cells	Δ	cells
0.00–0.25		24	+6.3	24
0.25–0.50		6	+8.5	6
0.50–0.75		6	+15.5	24
0.75–0.95		11	+4.2	33
0.95–1.00		34	−9.0	186
$r(\Delta, \text{baseline})$		−0.417		−0.050

Table 7: Mean primer effect (percentage points), binned by `none` accuracy on the same (arm, task, density, condition) cell of the main sweep, using the $n=40$ route split (Table 1). Mean baseline is 0.61 for route cells and 0.85 for no-route cells, so the groups are not baseline-matched; the cross-fitted interaction test in the text addresses this.

Estimate	r_{route}	k	r_{none}	k	interaction [95% CI]
naive, pp	−0.35	117	−0.01	381	−12.5 [−19.5, −6.3]
cross-fit (one direction), pp	−0.34	116	+0.03	381	−12.1 [−17.9, −7.0]
cross-fit (both), pp	−0.31	232	+0.02	762	−11.0 [−17.1, −5.7]
cross-fit (both), logit	−0.40	232	+0.04	762	−1.35 [−1.93, −0.77]
baseline .10–.90, pp	−0.01	78	−0.07	187	+3.0 [−29.1, +41.1]
baseline .10–.90, logit	+0.02	78	−0.05	187	+0.22 [−1.36, +1.94]
held-out, pp	−0.67	105	−0.19	93	−24.7 [−109, −3.8]

Table 8: Route $\times$ baseline relation, main sweep plus high-density extension (held-out: published split and probe). “Both” counts each cell once per fold direction; k for the window rows counts points. Interaction CIs resample (arm, task, density) blocks, or (arm, task) for held-out arms.

Comparison	Δ	n	density	p
clust. vs. none (main)	+3.0	400	.10–.50	0.26
clust. vs. none	+1.9	2800	.10–.85	0.022
filler vs. none	−2.4	2800	.10–.85	0.0013
clust. vs. filler	+4.3	2800	.10–.85	2.4×10^{-8}
clust. vs. none	+7.5	400	.35	0.0089
clust. vs. none (new seed)	+4.2	1600	.10–.50	0.00055

Table 9: qwen3-1.7b, `node_degree`. All rows except the first come from the dedicated 400-graph-per-density sweep; the last row uses newly seeded graphs. Per-density values are in Appendix Table 10.

p	clust.–none	filler–none	clust.–filler
.10	+0.3	+3.0*	−2.8*
.20	+3.3	+0.5	+2.8
.35	+7.5*	+1.0	+6.5*
.50	+4.3	−2.0	+6.3*
.65	−2.0	−9.5*	+7.5*
.75	+0.5	−6.3*	+6.8*
.85	−0.5	−3.5*	+3.0*

Table 10: qwen3-1.7b, `node_degree`, dedicated sweep, 400 graphs per density. *: McNemar $p < 0.05$ (uncorrected).

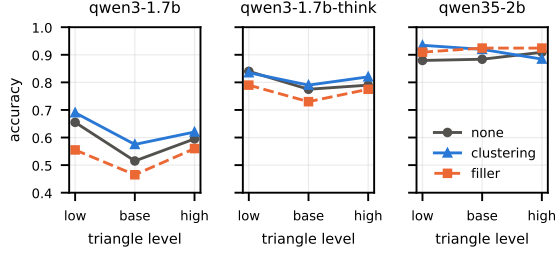


Figure 4: Rewiring: `node_degree` accuracy by triangle level (200 graphs per level; degree sequence and prompt length fixed; mean clustering 0.01, 0.31, 0.73).

Arm	low	base	high
predicted shift	-11.5	0.0	+16.7
qwen3-1.7b	+0.03	+0.28	-0.02
qwen3-1.7b-think	+0.01	+0.04	+0.01
qwen35-2b	-0.29	-0.26	-0.16

Table 11: Density-prior test on the rewiring corpus: mean paired shift in signed `node_degree` error, clustering minus none (errors clipped at ± 20 , 200 graphs per level). “Predicted” is the shift if answers moved fully to $\bar{C}$ ($n - 1$).

Task	1.7B, acc. under none (%)			4B, acc. under none (%)		
	plain	think	think@2048	plain	think	think@2048
connected_nodes	72.5	78.7	67.2	86.0	95.5	89.0
cycle_check	90.7	97.0	38.0	99.0	99.3	62.0
edge_count	1.3	15.8	0.0	2.0	20.0	0.0
edge_existence	70.5	99.0	97.8	90.5	99.8	99.8
node_count	31.5	91.0	86.0	90.2	99.5	94.8
node_degree	60.5	76.2	67.7	99.3	97.0	91.2

Table 12: Budget-matched comparison. Plain arms use 2,048 tokens; thinking arms 8,192 (“think”) or rescored at 2,048 (“think@2048”). Truncated generations are scored incorrect here, so values differ from the dropped-pair convention of Table 2.

Arm	Task	none	comp.	clust.	rwse	degree	filler	all
1.7B	node_count	0.69	0.47	0.01	0.67	0.62	0.29	0.22
	edge_count	133.4	98.1	105.7	79.0	47.7	181.2	94.2
	node_degree	0.94	0.91	0.78	0.91	0.67	1.27	0.91
4B	node_count	0.10	0.01	0.01	0.01	0.02	0.01	0.00
	edge_count	39.2	40.1	43.4	56.0	40.4	29.3	28.1
	node_degree	0.02	0.01	0.02	0.05	0.14	0.02	0.49

Table 13: Mean absolute error on the integer tasks, plain arms, main sweep (non-truncated rows).

A.3 Per-cell results

Paired effect sizes for every (arm, task, primer) cell: Δ in percentage points against none, a 95% paired-bootstrap interval, the permutation p -value, and Δ under the alternative convention that scores

truncated generations incorrect rather than dropping the pair (Section 3.4). n is the number of pairs in which neither arm truncated.

Task	Primer	n	Δ	95% CI	perm p	zeroed
connected_nodes	components	400	+0.8	[-2.8, +4.5]	0.7861	+0.8
	clustering	400	+1.2	[-3.0, +5.8]	0.6458	+1.2
	rwse	400	+1.0	[-3.5, +5.2]	0.7304	+1.0
	degree	400	+3.5	[-0.5, +7.5]	0.1239	+3.5
	filler	400	-6.2	[-10.2, -2.2]	0.0040	-6.2
	all	399	+2.0	[-2.8, +6.8]	0.4644	+2.0
cycle_check	components	343	-0.6	[-1.5, +0.0]	0.5033	+2.0
	clustering	362	+0.3	[+0.0, +0.8]	1.0000	+8.5
	rwse	358	-7.5	[-10.3, -4.7]	0.0001	-0.2
	degree	301	-12.6	[-16.6, -9.0]	0.0001	-17.5
	filler	341	+0.3	[+0.0, +0.9]	1.0000	+3.0
	all	362	-1.1	[-2.5, +0.0]	0.2210	+7.0
edge_count	components	211	-1.4	[-3.8, +0.5]	0.3783	-1.0
	clustering	238	-0.4	[-2.5, +1.7]	1.0000	-0.5
	rwse	239	-0.8	[-2.9, +1.3]	0.6837	-0.8
	degree	253	+4.0	[+0.8, +7.1]	0.0179	+3.0
	filler	224	-0.4	[-2.2, +1.3]	1.0000	-0.5
	all	211	+2.4	[-0.5, +5.7]	0.2269	+0.8
edge_existence	components	400	-8.2	[-11.8, -4.8]	0.0001	-8.2
	clustering	400	+3.8	[-0.5, +8.0]	0.1199	+3.8
	rwse	399	+5.3	[+0.8, +9.8]	0.0321	+5.0
	degree	400	+4.0	[-0.5, +8.2]	0.0957	+4.0
	filler	400	-15.8	[-20.0, -11.8]	0.0001	-15.8
	all	400	+11.5	[+6.8, +16.2]	0.0001	+11.5
node_count	components	400	+21.2	[+15.8, +26.8]	0.0001	+21.2
	clustering	400	+67.2	[+62.7, +71.8]	0.0001	+67.2
	rwse	400	+2.0	[-4.2, +8.5]	0.5871	+2.0
	degree	400	+6.8	[+3.0, +10.5]	0.0008	+6.8
	filler	400	+39.8	[+35.0, +44.8]	0.0001	+39.8
	all	400	+47.0	[+42.0, +52.0]	0.0001	+47.0
node_degree	components	400	+1.8	[-2.0, +5.8]	0.4599	+1.8
	clustering	400	+3.0	[-1.8, +7.8]	0.2577	+3.0
	rwse	399	+4.0	[-0.5, +8.5]	0.1104	+4.0
	degree	400	+7.5	[+3.0, +12.0]	0.0018	+7.5
	filler	400	-0.5	[-4.5, +3.5]	0.9028	-0.5
	all	400	+10.2	[+5.2, +15.2]	0.0003	+10.2

Table 14: qwen3-1.7b: all cells.

Task	Primer	n	Δ	95% CI	perm p	zeroed
connected_nodes	components	392	+2.8	[-0.5, +6.1]	0.1268	+2.5
	clustering	394	+1.5	[-2.5, +5.6]	0.5384	+2.0
	rwse	393	-0.3	[-4.1, +3.6]	1.0000	+0.2
	degree	390	+7.4	[+3.6, +11.3]	0.0001	+6.2
	filler	394	+2.0	[-1.5, +5.6]	0.3275	+2.5
	all	392	+6.1	[+2.6, +9.9]	0.0027	+6.2
cycle_check	components	346	+0.0	[+0.0, +0.0]	1.0000	-7.5
	clustering	376	+0.0	[+0.0, +0.0]	1.0000	-0.5
	rwse	386	+0.0	[+0.0, +0.0]	1.0000	+2.2
	degree	295	-0.7	[-1.7, +0.0]	0.5036	-21.8
	filler	388	+0.0	[+0.0, +0.0]	1.0000	+3.0
	all	373	+0.0	[+0.0, +0.0]	1.0000	-0.8
edge_count	components	27	+7.4	[+0.0, +18.5]	0.5037	-0.8
	clustering	23	-8.7	[-26.1, +8.7]	0.6251	-1.2
	rwse	25	-4.0	[-12.0, +0.0]	1.0000	-3.2
	degree	13	-61.5	[-84.6, -38.5]	0.0102	-3.2
	filler	29	+3.4	[-6.9, +13.8]	1.0000	-1.5
	all	24	-8.3	[-29.2, +12.5]	0.6794	+5.0
edge_existence	components	398	-0.8	[-2.3, +0.8]	0.5090	-0.8
	clustering	398	-1.0	[-2.5, +0.3]	0.2911	-1.0
	rwse	398	-1.3	[-3.0, +0.3]	0.2274	-1.2
	degree	397	-2.3	[-4.3, -0.5]	0.0249	-2.5
	filler	399	-0.8	[-2.3, +0.8]	0.5051	-0.5
	all	399	-2.0	[-3.8, -0.5]	0.0374	-1.8
node_count	components	342	+0.9	[-0.3, +2.3]	0.3758	+1.5
	clustering	369	+1.4	[+0.3, +2.7]	0.0597	+9.0
	rwse	369	+1.4	[+0.3, +2.7]	0.0597	+9.0
	degree	360	+0.8	[-0.6, +2.2]	0.4512	+5.5
	filler	369	+1.4	[+0.3, +2.7]	0.0597	+9.0
	all	369	+1.4	[+0.3, +2.7]	0.0597	+9.0
node_degree	components	395	-1.0	[-4.3, +2.3]	0.6543	-1.0
	clustering	396	+1.0	[-2.8, +4.8]	0.7010	+1.5
	rwse	394	+2.5	[-1.5, +6.9]	0.2898	+2.5
	degree	376	+11.7	[+7.4, +16.0]	0.0001	+8.8
	filler	396	-3.5	[-7.3, +0.3]	0.0954	-3.0
	all	396	+11.6	[+7.3, +15.9]	0.0001	+12.2

Table 15: qwen3-1.7b-think: all cells.

Task	Primer	n	Δ	95% CI	perm p	zeroed
connected_nodes	components	400	+9.2	[+6.0, +12.8]	0.0001	+9.2
	clustering	400	+1.0	[-3.0, +4.8]	0.6893	+1.0
	rwse	400	+0.0	[-4.0, +4.0]	1.0000	+0.0
	degree	400	-3.0	[-7.2, +1.2]	0.1882	-3.0
	filler	400	-12.5	[-17.0, -8.0]	0.0001	-12.5
	all	400	-2.8	[-7.0, +1.5]	0.2511	-2.8
cycle_check	components	391	-1.3	[-2.6, +0.0]	0.1318	-2.0
	clustering	395	+0.3	[+0.0, +0.8]	1.0000	+0.5
	rwse	392	+0.3	[+0.0, +0.8]	1.0000	-0.2
	degree	391	-1.5	[-2.8, -0.5]	0.0312	-2.2
	filler	392	-0.3	[-0.8, +0.0]	1.0000	-0.5
	all	395	-0.5	[-1.5, +0.5]	0.6269	-0.2
edge_count	components	397	-1.3	[-2.8, +0.3]	0.1756	-1.2
	clustering	399	-1.3	[-3.0, +0.3]	0.2298	-1.2
	rwse	398	-1.5	[-3.0, -0.3]	0.0668	-1.5
	degree	393	+27.7	[+23.2, +32.6]	0.0001	+27.3
	filler	400	+1.2	[-0.5, +3.2]	0.3012	+1.2
	all	397	+3.8	[+1.3, +6.3]	0.0049	+3.8
edge_existence	components	400	-2.8	[-6.0, +0.5]	0.1254	-2.8
	clustering	400	+4.0	[+1.0, +7.0]	0.0168	+4.0
	rwse	400	-10.0	[-14.0, -6.2]	0.0001	-10.0
	degree	398	-4.3	[-7.8, -1.0]	0.0200	-4.8
	filler	400	-7.0	[-10.8, -3.5]	0.0002	-7.0
	all	399	-9.8	[-13.8, -6.0]	0.0001	-10.0
node_count	components	400	+9.0	[+6.2, +12.0]	0.0001	+9.0
	clustering	400	+8.8	[+5.8, +11.8]	0.0001	+8.8
	rwse	400	+9.2	[+6.2, +12.2]	0.0001	+9.2
	degree	400	+8.0	[+5.0, +11.2]	0.0001	+8.0
	filler	400	+9.2	[+6.5, +12.2]	0.0001	+9.2
	all	400	+9.8	[+7.0, +12.8]	0.0001	+9.8
node_degree	components	400	+0.2	[+0.0, +0.8]	1.0000	+0.2
	clustering	400	-0.5	[-2.0, +0.8]	0.7246	-0.5
	rwse	400	-2.2	[-4.0, -0.5]	0.0240	-2.2
	degree	400	-6.5	[-9.2, -3.8]	0.0001	-6.5
	filler	400	-0.5	[-2.0, +0.8]	0.7232	-0.5
	all	400	-8.0	[-10.8, -5.2]	0.0001	-8.0

Table 16: qwen3-4b: all cells.

Task	Primer	n	Δ	95% CI	perm p	zeroed
connected_nodes	components	377	+0.5	[-0.5, +1.6]	0.6249	+0.5
	clustering	373	+0.0	[-1.3, +1.3]	1.0000	-0.8
	rwse	379	+0.0	[-1.3, +1.6]	1.0000	+0.5
	degree	376	+0.0	[-1.6, +1.6]	1.0000	+0.5
	filler	379	+0.8	[-0.3, +2.1]	0.3787	+1.2
	all	375	-0.5	[-2.1, +0.8]	0.7306	-1.2
cycle_check	components	381	+0.0	[+0.0, +0.0]	1.0000	-3.2
	clustering	394	+0.0	[+0.0, +0.0]	1.0000	+0.0
	rwse	395	+0.0	[+0.0, +0.0]	1.0000	+0.2
	degree	375	-0.8	[-1.9, +0.0]	0.2514	-5.5
	filler	396	+0.0	[+0.0, +0.0]	1.0000	+0.5
	all	392	+0.0	[+0.0, +0.0]	1.0000	-0.5
edge_count	components	60	+1.7	[+0.0, +5.0]	1.0000	+24.8
	clustering	59	+1.7	[-3.4, +6.8]	1.0000	+24.8
	rwse	49	+0.0	[-6.1, +6.1]	1.0000	+10.2
	degree	34	-23.5	[-38.2, -8.8]	0.0073	+8.5
	filler	50	-2.0	[-6.0, +0.0]	1.0000	+12.2
	all	41	-41.5	[-56.1, -26.8]	0.0002	+1.8
edge_existence	components	396	+0.0	[+0.0, +0.0]	1.0000	-0.5
	clustering	396	+0.0	[+0.0, +0.0]	1.0000	-0.8
	rwse	393	+0.0	[+0.0, +0.0]	1.0000	-1.2
	degree	398	+0.0	[+0.0, +0.0]	1.0000	+0.0
	filler	399	+0.0	[+0.0, +0.0]	1.0000	+0.0
	all	394	+0.0	[+0.0, +0.0]	1.0000	-1.0
node_count	components	400	+0.0	[-1.0, +1.0]	1.0000	+0.0
	clustering	399	+0.5	[+0.0, +1.3]	0.4936	+0.2
	rwse	399	+0.5	[+0.0, +1.3]	0.5005	+0.2
	degree	399	+0.5	[+0.0, +1.3]	0.5005	+0.2
	filler	400	+0.5	[+0.0, +1.2]	0.4870	+0.5
	all	400	+0.5	[+0.0, +1.2]	0.4870	+0.5
node_degree	components	380	-0.3	[-0.8, +0.0]	1.0000	+0.2
	clustering	386	-0.3	[-1.0, +0.5]	1.0000	+1.8
	rwse	381	-2.1	[-3.7, -0.8]	0.0072	-1.2
	degree	388	+0.0	[-0.8, +0.8]	1.0000	+2.5
	filler	386	+0.0	[+0.0, +0.0]	1.0000	+2.0
	all	383	-0.8	[-1.8, +0.0]	0.2466	+0.5

Table 17: qwen3-4b-think: all cells.